

Rice Mill Data Tracking — Replacing Airtable

September 17, 2026

The problem

Field operators record production data (bags milled, revenue, expenses) in Airtable, but the on-site internet connection is unreliable.

- As a result, sheets aren't filled in regularly — there are gaps and delays of several days.
- The owner needs these figures in his Excel tracking file as fast as possible after each entry.
- Any solution must therefore work without a connection, while keeping data reliable once submitted.

Goals and constraints

1. **Work offline:** data entry possible without a network, automatic sync once the connection returns.
2. **Data integrity:** once submitted, an entry should no longer be freely editable by the field operator.
3. **Speed to Excel:** validated data should reach the owner's file as fast as possible, ideally within minutes.
4. **Ease of use:** lightweight app, installable on mobile, no complicated account management in the field.

Options considered

Option	Offline reliability	Data control	Speed to Excel	Effort
Stay on Airtable	Low — blocked without network	Low — anyone can edit anything	Medium — manual export/sync	None
Google Sheets + third-party app	Low to medium	Low	Medium	Low
Custom offline-first PWA (recommended)	High — local entry, auto sync	High — locked after submission + audit trail	High — automatic push	High — custom development

Recommended solution: offline-first PWA

A progressive web app (PWA), installable on the operator's phone, with this flow:

1. **Local entry:** the form (bags milled, revenue, expenses, other) writes first to the phone's local storage (IndexedDB) — works even without a network.
2. **Automatic sync:** as soon as a connection is detected, pending entries are sent to the server in the background, with no action from the user.

3. **Locked after submission:** once synced, an entry becomes read-only for the operator. A correction is only possible through a tracked request (reason + edit history).
4. **Push to Excel:** as soon as an entry is confirmed server-side, it is automatically pushed to the owner's file (e.g. via the Microsoft Graph API / Power Automate if Excel lives on OneDrive-SharePoint, or the Google Sheets API if the owner accepts a connected Sheet). Target delay: a few minutes after sync, not a manual end-of-day export.

This architecture directly addresses the three constraints: field reliability (local storage), integrity (lock + audit trail), and speed (automatic push instead of manual entry/export).

Next steps

- Confirm the exact list of fields to capture (bags milled, revenue, expenses, other categories)
- Decide on the owner's landing point (Excel on OneDrive/SharePoint vs. a connected Google Sheet)
- Build a PWA prototype (one form, local storage, sync)
- Test under real conditions with a client who has an unstable connection
- Implement entry locking and the edit history